

Windows to Linux Translation Guide

Filesystem & Paths · RH124 Companion Reference

1. The Mental Model Shift — One Tree, Not Many Drives

The biggest conceptual hurdle for Windows admins: Linux has a single directory tree rooted at `/`. There are no drive letters. Storage devices, USB sticks, and CD-ROMs are all *mounted* as folders within that single tree.

Windows	Linux
C:\, D:\, E:\ — multiple drive letters	<code>/</code> — one single tree, everything hangs off it
CD-ROM appears as D:\	CD-ROM mounted at <code>/mnt/cdrom</code> or <code>/run/media/...</code>
USB drive appears as E:\	USB drive appears at <code>/run/media/username/label</code>
Drives managed by Disk Management	Drives managed with <code>mount</code> / <code>umount</code> commands

"In Linux, drives do not get letters — they get folders. When you plug in a USB stick, Linux attaches it to a folder in the tree. The tree itself never changes shape."

2. Path Separator — The Slash Is Flipped

Windows	Linux
C:\Users\john\Documents	<code>/home/john/documents</code>
Backslash <code>\</code> as separator	Forward slash <code>/</code> as separator
Case-insensitive (NTFS default)	Case-sensitive — always

*The backslash `\` in Linux is an **escape character**, not a path separator. Also: **Document.txt**, **document.txt**, and **DOCUMENT.TXT** are three completely different files on Linux. This catches every Windows person at least once.*

3. Directory Equivalents

Windows	Linux Equivalent	Notes
C:\	<code>/</code>	Root of everything
C:\Windows\System32	<code>/usr/bin</code> , <code>/usr/sbin</code>	System executables
C:\Windows\System32\drivers\etc\hosts	<code>/etc/hosts</code>	Hosts file — same purpose
C:\Program Files	<code>/usr/local</code> or <code>/opt</code>	Installed applications
C:\Users\john	<code>/home/john</code>	User home directory
C:\Users\Administrator	<code>/root</code>	Admin/superuser home
C:\Windows\Temp	<code>/tmp</code>	Temporary files (cleared on reboot)
C:\Windows	<code>/usr</code>	OS files and libraries
C:\Windows\System32\config	<code>/etc</code>	System configuration (plain text files)
C:\Windows\Logs / Event Viewer	<code>/var/log</code>	System logs
Device Manager	<code>/dev</code>	Device files
%APPDATA%	<code>~/.config</code> or <code>~/.local</code>	User application data

Registry	/etc (plain text files)	No registry — everything is readable text
Task Manager	top, ps, htop	Process management
Services (services.msc)	systemctl	Service management
Event Viewer	journalctl	Log viewer

4. Hidden Files — No Hidden Attribute, Just a Dot

Windows	Linux
Right-click → Properties → Hidden attribute	Filename starts with a . (dot)
Hidden via filesystem metadata	Hidden by naming convention only
Show hidden files in Explorer options	ls -a shows all files including dotfiles

User config files like **.bashrc**, **.ssh**, and **.gitconfig** are hidden by default so they do not clutter **ls** output. Use **ls -la** to see everything with full details.

5. Security: Current Directory Is Not in PATH

Windows	Linux
Typing script.bat runs it if in current dir	Typing script.sh gives 'command not found'
Current dir implicitly searched in PATH	Must explicitly type ./script.sh to run
Risk: malicious binary in current dir runs accidentally	Protection: accidental execution prevented by design

Linux deliberately excludes . (the current directory) from the default PATH. You must type **./myscript.sh** to run a script in the current directory. This prevents accidentally running a malicious binary that someone dropped in your working directory.

6. Useful Path Shortcuts

Shortcut	Meaning	Example
~	Current user's home directory	cd ~ or just cd
~user	Another user's home directory	ls ~root
.	Current directory	cp /etc/hosts .
..	Parent directory	cd ..
-	Previous working directory	cd - (like a Back button)
\$PWD	Environment variable: current directory	echo \$PWD
\$HOME	Environment variable: home directory	echo \$HOME

7. The Registry vs Plain Text Config Files

Windows	Linux
Binary registry hive — needs regedit	Plain text files in /etc — any editor works
Corruption can brick the OS	Bad config file is fixable with a text editor
Hard to diff, version-control, or audit	Can be diffed, backed up, git-tracked easily
Requires GUI or reg.exe to inspect	cat /etc/ssh/sshd_config — human-readable
Organised in keys/values/hives	Organised as files and directories

"Everything is a text file. If you can read it, you can understand it. If you can understand it, you can fix it."

8. /tmp vs /var/tmp — A Security-Relevant Distinction

Windows Equivalent	Linux
C:\Windows\Temp — persists until manually cleared	/tmp — cleared on every reboot, files auto-purged after 10 days
No equivalent	/var/tmp — persists across reboots, purged after 30 days
NTFS permissions restrict who can write	World-writable by default — anyone can write files

Security note: Because /tmp is world-writable, it is a common target for staging malware. Security hardening guides recommend mounting /tmp with **noexec** and **nosuid** mount options to prevent execution of binaries placed there.